



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SRI VENKATASWARA SERVICE STN.

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

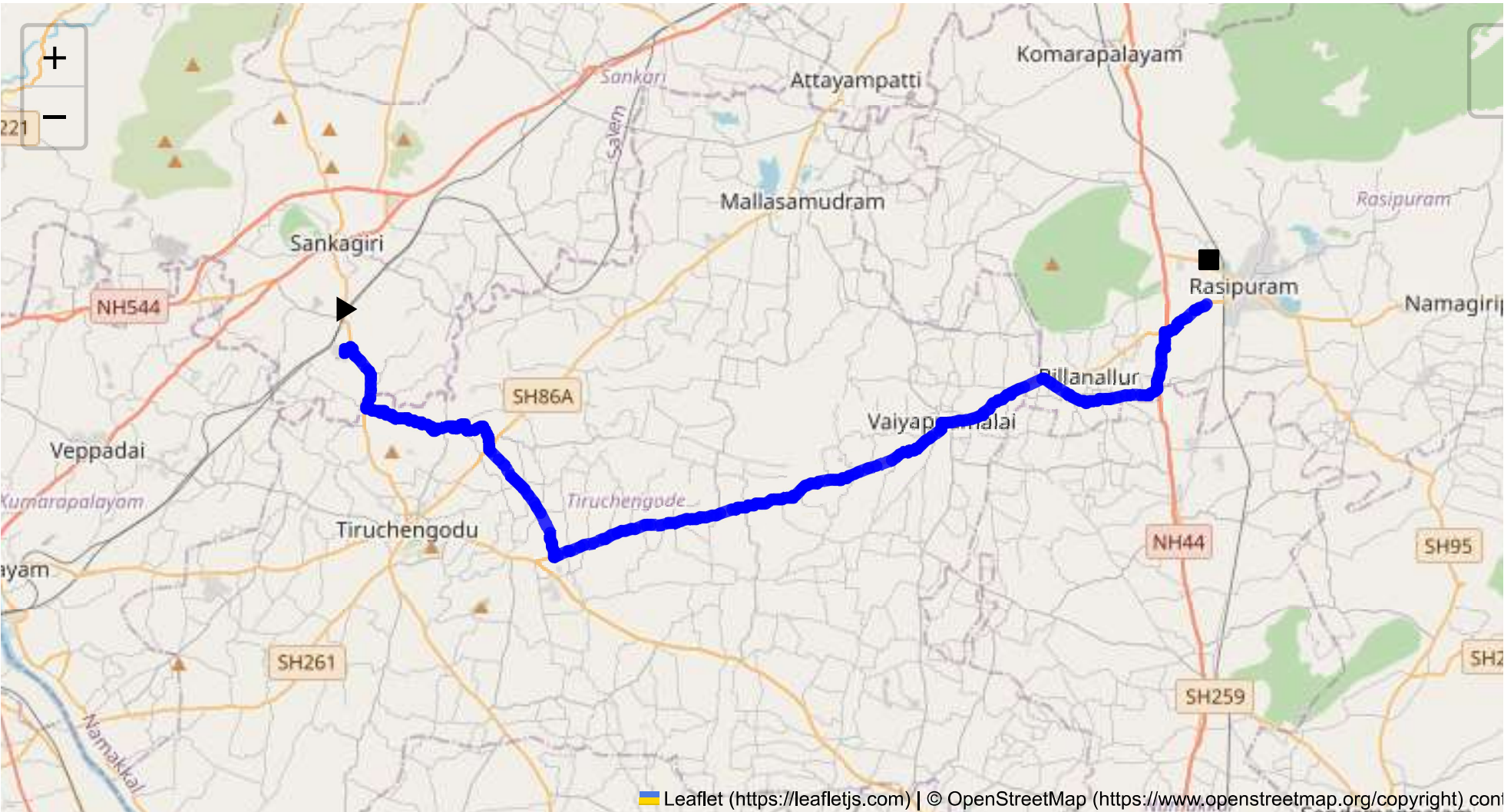
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 42.70 km
Estimated Duration: 1.0 hours
Adjusted Duration (Heavy Vehicle): 1.2 hours
Start: (11.4381, 77.8734)
End: (11.455126, 78.172432)



Welcome to the Journey Risk Management Study

Route Safety Analysis: Sangagiri to Rasipuram, Tamil Nadu

1. Overview of the Route Map:

- The route from Sangagiri to Rasipuram covers approximately 42.7 km and involves navigating through NH79 and SH79, primarily local roads with occasional highways. The journey typically takes about 58 minutes for heavy vehicles under optimal conditions.

2. Typical Weather Conditions and Potential Hazards:

- The region experiences a tropical climate with relatively high temperatures year-round. The monsoon season (June to September) can bring heavy rains, causing potential flooding and slippery road conditions. Fog can occur in the early mornings during the winter months (November to February), reducing visibility.

3. Traffic Patterns and Congestion:

- Traffic is generally moderate, with peak hours likely between 8:00 to 10:00 AM and 5:00 to 7:00 PM. Congestion may occur around busy market areas and near major junctions, particularly in towns like Rasipuram.

4. Road Quality and Infrastructure:

- The highways are generally in good condition, but some local roads may have potholes or uneven surfaces. There are ongoing roadworks in certain sections, which could affect travel times. Bridges and tunnels are usually well-maintained, though caution is advised during the rainy season due to potential structural vulnerabilities.

5. Alternative Routes for Emergencies:

- In case of emergencies, consider taking NH544 (Salem to Kumarapalayam) as an alternative before connecting to NH79 towards Rasipuram. This detour might add distance but typically offers better road quality and fewer hazards.

6. Local Regulations on Hazardous Material Transport:

- Transporting hazardous materials requires compliance with local and national safety standards. This includes properly secured cargo, emergency response plans, and adherence to speed limits and loading regulations.

7. Historical Incidents:

- There have been occasional incidents involving heavy vehicles, primarily due to driver fatigue or adverse weather conditions. Awareness and adherence to safety protocols are essential to mitigate such risks.

8. Environmental Considerations:

- The route passes near agricultural lands and small communities. Drivers should minimize noise and emissions, with emphasis on conservation practices. Awareness of local wildlife crossings and habitats is advised.

9. Communication Coverage:

- Mobile network coverage is generally reliable along the main highways but may be spotty in more rural areas or near dense vegetation.

10. Emergency Response Times:

- Urban and semi-urban areas may experience response times of approximately 15-30 minutes, while rural segments may see delays of up to an hour due to distance from key emergency services.

11. Overall Summary of Risk Assessment:

- The route is moderately safe under typical conditions but requires caution during adverse weather or peak traffic hours. Ensure proper vehicle maintenance and compliance with local regulations to minimize risks. Maintain communication with the dispatch center, especially in areas of identified network gaps, and always have an emergency plan prepared.

Preparation, adherence to regulations, and situational awareness are key to safely navigating this route for truck drivers, especially when carrying hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
3	Blind Spot	Blind Spot	11.41975, 77.88075	10 KM/Hr
4	Turn	Medium	11.41876, 77.88487	30 KM/Hr
5	Turn	Medium	11.41829, 77.88568	30 KM/Hr
6	Turn	High	11.41859, 77.88692	15 KM/Hr
7	Turn	Medium	11.41761, 77.88739	30 KM/Hr
8	Turn	Medium	11.41716, 77.88935	30 KM/Hr
9	Turn	Medium	11.41630, 77.89077	30 KM/Hr
10	Turn	Medium	11.41601, 77.89198	30 KM/Hr
11	Turn	Medium	11.41603, 77.89479	30 KM/Hr
12	Turn	Medium	11.41439, 77.90025	30 KM/Hr
13	Turn	High	11.41377, 77.90073	15 KM/Hr
14	Turn	Medium	11.41375, 77.90128	30 KM/Hr
15	Turn	Medium	11.41389, 77.90161	30 KM/Hr
16	Turn	Medium	11.41325, 77.90313	30 KM/Hr
17	Turn	Medium	11.41276, 77.90349	30 KM/Hr
18	Turn	Medium	11.41227, 77.90470	30 KM/Hr
19	Turn	High	11.41230, 77.90485	15 KM/Hr
20	Turn	Medium	11.41262, 77.90506	30 KM/Hr
21	Turn	Medium	11.41273, 77.90524	30 KM/Hr
22	Turn	High	11.41329, 77.90978	15 KM/Hr
23	Turn	Medium	11.41314, 77.91006	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
24	Turn	High	11.41313, 77.91057	15 KM/Hr
25	Turn	High	11.41307, 77.91058	15 KM/Hr
26	Turn	Medium	11.41334, 77.91266	30 KM/Hr
27	Turn	High	11.41424, 77.91433	15 KM/Hr
28	Turn	Medium	11.41409, 77.91458	30 KM/Hr
29	Turn	High	11.41208, 77.91551	15 KM/Hr
30	Turn	High	11.41347, 77.92096	15 KM/Hr
31	Turn	Medium	11.41336, 77.92130	30 KM/Hr
32	Turn	Medium	11.40833, 77.92362	30 KM/Hr
33	Turn	High	11.40566, 77.92355	15 KM/Hr
34	Turn	High	11.40557, 77.92343	15 KM/Hr
35	Blind Spot	Blind Spot	11.36920, 77.94568	10 KM/Hr
36	Turn	High	11.41466, 78.08070	15 KM/Hr
37	Turn	Medium	11.41473, 78.08082	30 KM/Hr
38	Turn	High	11.42964, 78.11587	15 KM/Hr
39	Turn	Medium	11.42956, 78.11590	30 KM/Hr
40	Turn	Medium	11.42450, 78.15278	30 KM/Hr
41	Turn	High	11.42622, 78.15562	15 KM/Hr
42	Turn	High	11.44547, 78.15837	15 KM/Hr

Emergency Locations

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	marketplace	Farmer Market	11.4146849, 78.0805463	30 km/h	Medium
1	school	Government Higher Secondary School	11.4150045, 78.0838172	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
2	school	Government school	11.4248027, 78.1323902	30 km/h	Medium
3	college	Paavai Arts And Science College For Women	11.4262443, 78.1546987	30 km/h	Medium
4	school	SRV EXCEL Matriculation Higher Secondary School	11.4299679, 78.152616	30 km/h	Medium
5	college	Thiruvalluvar Govt arts college	11.4449352, 78.1595562	30 km/h	Medium
6	school	Venkateswara School	11.4468258, 78.1595761	30 km/h	Medium
7	school	Venkateswara Middle school	11.4472063, 78.1596191	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.41975, 77.88075



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41876, 77.88487



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41829, 77.88568



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41859, 77.88692



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41761, 77.88739



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41716, 77.88935



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41630, 77.89077



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41601, 77.89198



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41603, 77.89479



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41439, 77.90025



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41377, 77.90073



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41375, 77.90128



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41389, 77.90161



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41325, 77.90313



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41276, 77.90349



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41227, 77.90470



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41230, 77.90485



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41262, 77.90506



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41273, 77.90524



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41329, 77.90978



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41314, 77.91006



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.41313, 77.91057



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41307, 77.91058



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41334, 77.91266



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41424, 77.91433



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41409, 77.91458



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41208, 77.91551



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41347, 77.92096



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41336, 77.92130



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.40833, 77.92362



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.40566, 77.92355



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.40557, 77.92343



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.36920, 77.94568



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41466, 78.08070



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41473, 78.08082



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42964, 78.11587



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42956, 78.11590



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42450, 78.15278



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42622, 78.15562



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44547, 78.15837

